

2-level Models for Individual Change



Background: Problems in Measuring Change

◆ Goals

- Assess each child's rate of learning or change
- Predict future status (e.g., identify kids at risk)
- Compare growth rates (e.g., SES gaps in learning)
- Correlate initial status and rate of learning
- Adjust for prior differences in examining the impact of a treatment (e.g., “value added”).

Core Ideas

- ◆ Conceptualization
 - Formulate a model for individual growth
 - Compare people by comparing their growth parameters
- ◆ Measurement
 - Create an interval scale that is invariant over time
 - Study reliability of measures of *change*
- ◆ Design
 - Frequency and duration of observation
 - Sample size

Example: Cognitive Skill During Early Childhood

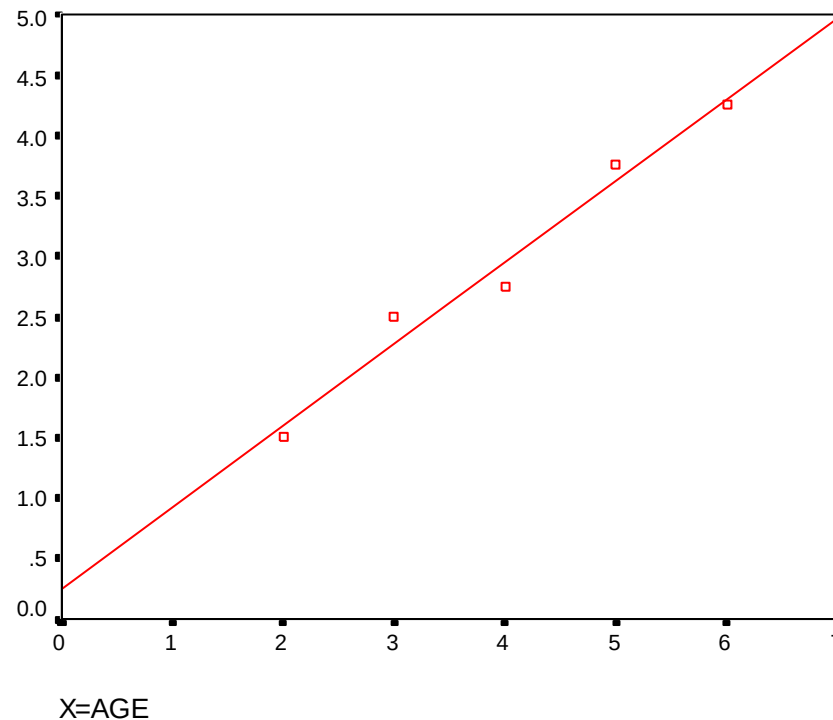
A. Conceptual Framework

- A model for one child
- A model for two children
- A model for J children
- Incorporating level-2 predictors

B. Illustration

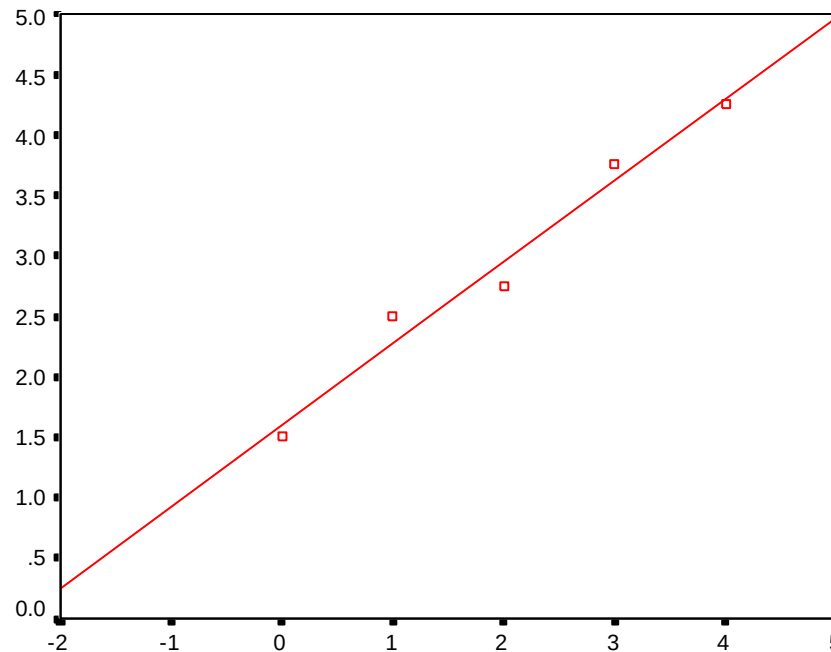
- Organization of the data at each level
- Building the combined data file
- Exploratory analysis
- An unconditional level-2 model
- Incorporating level-2 covariates
- Checking assumptions

Plot of Cognitive Skill As a Function of Age for One Child



$$Y_i = \pi_0 + \pi_1(\text{Age}_i) + e_i$$

Plot of Cognitive Skill As a Function of Age-2 for One Child

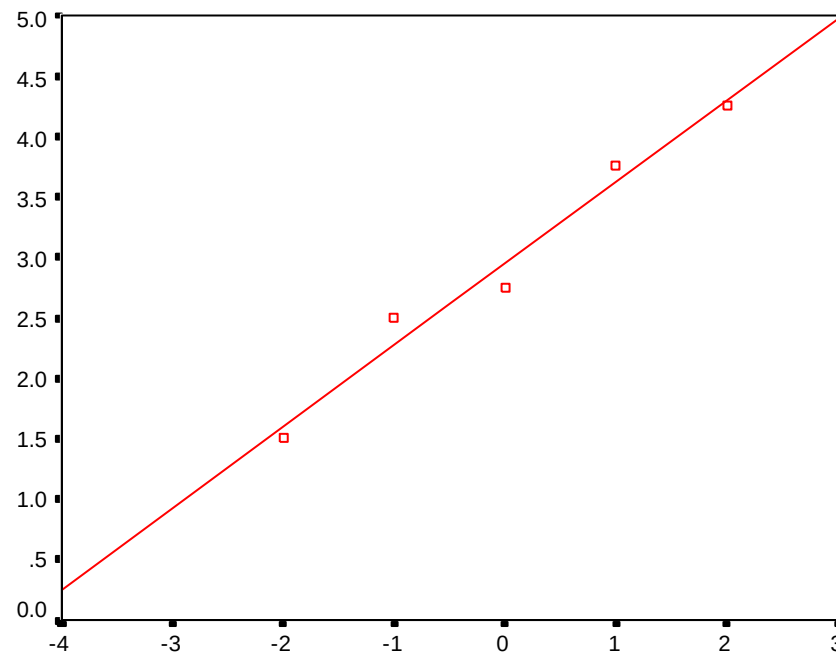


X=Age-2

$$Y_i = \pi_0 + \pi_1(\text{Age}_i - 2) + e_i$$

π_0 = status at entry (age 2)

Plot of Cognitive Skill As a Function of Age-4 for One Child

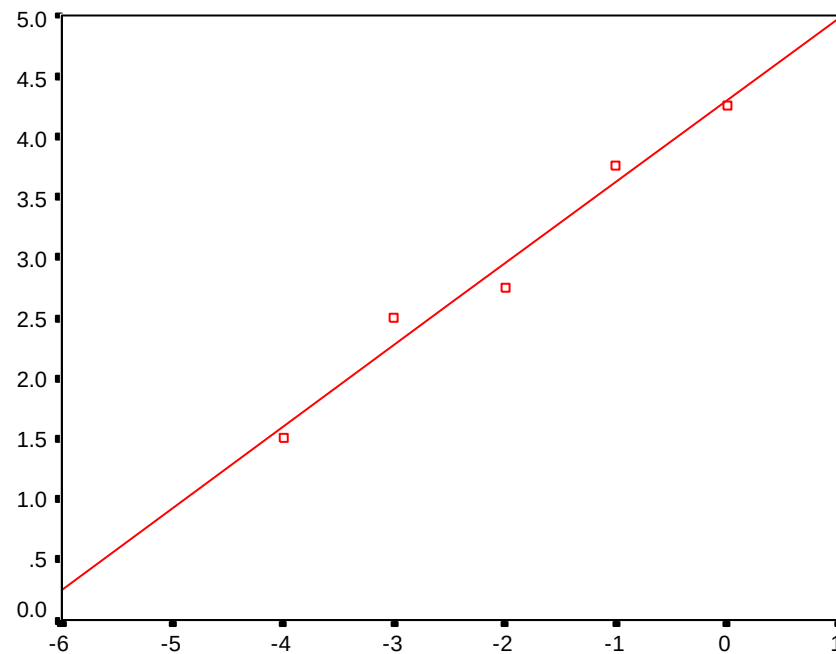


X=Age-4

$$Y_i = \pi_0 + \pi_1(\text{Age}_i - 4) + e_i$$

π_0 = status at age 4 = mean status

Now Using Age-6



X=Age-6

$$Y_i = \pi_0 + \pi_1(\text{Age}_i - 6) + e_i$$

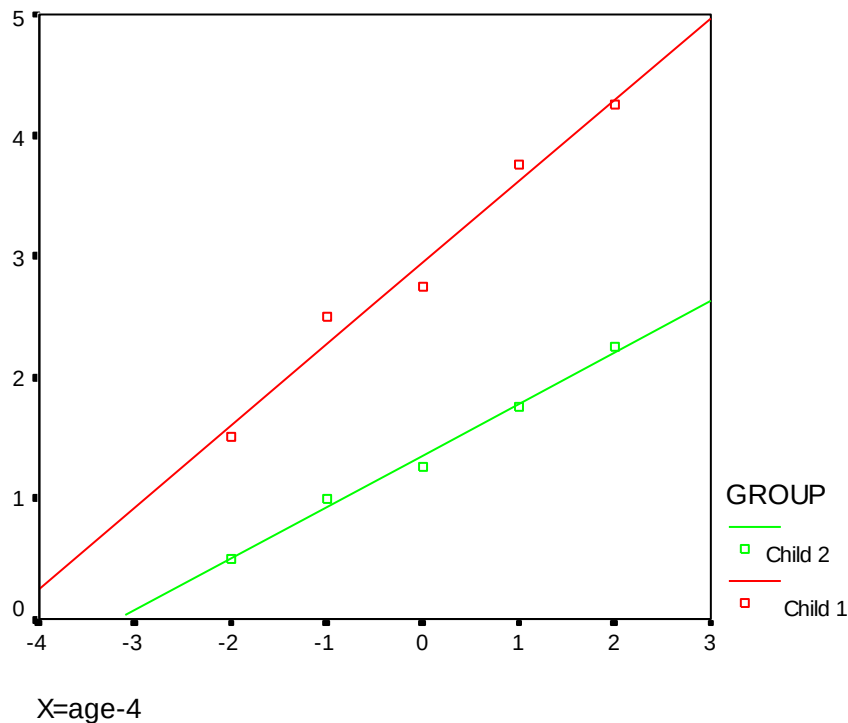
π_0 = status at age 6 = status at end point

A Favored Option

Centering around the middle of the trajectory is often best. It offers several distinct advantages

- Linear growth parameters represent growth rates that are often close to the average growth rate over the whole trajectory during the study period
- It improves estimation of the intercept by ensuring data are on both sides of it on the x-axis and eliminates potential extrapolation
- It breaks the correlation between linear and quadratic terms. This makes it easier to estimate these terms (resulting in narrower confidence intervals and increased reliability).

Cognitive Skill As a Function of Age-4 for Two Children



$$Y_{i1} = \pi_{01} + \pi_{11}(\text{Age}_{i1}-4) + e_{i1}$$

$$Y_{i2} = \pi_{02} + \pi_{12}(\text{Age}_{i2}-4) + e_{i2}$$

Note: $\pi_{02} < \pi_{01}$
 $\pi_{12} < \pi_{11}$

Study of Age-Outcome Relationship for J Children

$$Y_{ti} = \pi_{0i} + \pi_{1i}(\text{Age} - 4)_{ti} + e_{ti}$$

$$E(\pi_{0i}) = \beta_0 \quad \text{Var}(\pi_{0i}) = \tau_{00}$$

$$E(\pi_{1i}) = \beta_1 \quad \text{Var}(\pi_{1i}) = \tau_{11}$$

$$\text{Cov}(\pi_{0i}, \pi_{1i}) = \tau_{01}$$

Modeling Correlates of Growth

Within each child :

$$Y_{ti} = \pi_{0i} + \pi_{1i} (Age - 4)_{ti} + e_{ti}$$

Across children :

$$\pi_{0i} = \beta_{00} + \beta_{01} X_i + r_{0i}$$

$$\pi_{1i} = \beta_{10} + \beta_{11} X_i + r_{1i}$$

Combined Model :

$$Y_{ti} = \beta_{00} + \beta_{01} X_i + \beta_{10} (Age - 4)_{ti} + \beta_{11} X_i (Age - 4)_{ti} + r_{0i} + (Age - 4)_{ti} r_{1i} + e_{ti}$$

Typical Steps on Studying Change

- ◆ 1. Specifying the individual growth model
 - Exploring the data
 - Finding the simplest polynomial that adequately fits the data
 - Centering
- ◆ 2. Studying correlates of growth
 - Identifying key predictors
 - Determining the covariance structure
- ◆ 3. Checking assumptions
 - Normality of random coefficients
 - Homogeneity of variance at each level
 - Linearity
 - Omitted predictors

Quadratic model

Level – 1 Model

$$Y_{ti} = \pi_{0i} + \pi_{1i}(age_{ti} - 4) + \pi_{2i}(age_{ti} - 4)^2 + e_{ti}$$

$$E(Y_{ti} | age_{ti} = 4) = \pi_{0i}$$

Defining the linear term

Level -1 Model

$$Y_{ti} = \pi_{0i} + \pi_{1i}(age_{ti} - 4) + \pi_{2i}(age_{ti} - 4)^2 + e_{ti}$$

$$E(Y_{ti} | age_{ti} = 4) = \pi_{0i}$$

$$\frac{dY_{ti}}{d(Age_{ti} - 4)} = \pi_{1i} + 2\pi_{2i}(age_{ti} - 4)$$

$$E\left(\frac{dY_{ti}}{d(Age_{ti} - 4)} \mid age_{ti} = 4\right) = \pi_{1i}$$

Defining the linear term

